

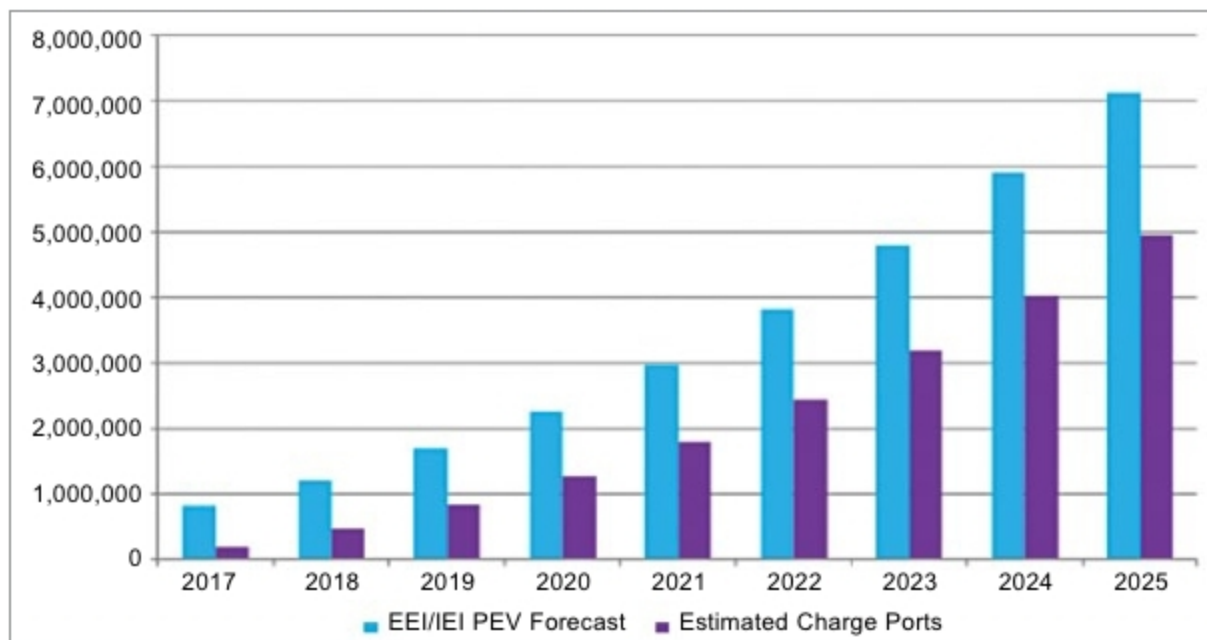


Fig. 2: Plug-in EV stock and charging infrastructure (charge ports) needed (2017-2025)
(Credit: Edison Electric Institute)

range, performance issues, long charge time and a dearth of charging stations will be wiped out.

Factors driving EV market

Growth of EV market is driven by government funding, subsidies and incentives, growing demand for EVs, increasing concerns over environmental pollution and huge investments from automakers in EVs. World Economic Forum recently released a report on the EV ecosystem stating India has the potential to become the largest EV market in the world. But the role of the government is crucial for accelerating adoption in addition to long-term investment in research and development (R&D) to create sustained growth.

Auto experts state that if the shift of transportation is moved to EVs, there will be less dependence on oil imports because a huge part of the oil is used for running traditional vehicles. It would allow the government to free up these funds towards establishment of the needed infrastructure to run EVs, such as charging stations across the country and battery production capacity.

Another benefit of an EV is its storage potential. If it is amalgamated with the ability to generate power using solar energy, EV batteries can help with power grid balancing.

Demand for EVs is driven by increased demand for fuel-efficient vehicles, high performance and low vehicle emission. Various issues such as vigorous developments in policies, implementation of guidelines and

advances in technology are stimulating the demand for EVs. Moreover, reduction in prices of lithium-ion batteries is likely to reduce the overall cost of producing EVs, which, in turn, will encourage more consumers to buy EVs.

Currently, EVs constitute less than one per cent of all the vehicles sold in India. There are more than 400,000 units of electric two-wheelers and only a few thousand electric cars on Indian roads. **Manufacturers are waiting for the government to clear regulatory hurdles and come up with a clear stance on infrastructure development for EVs. Subsidies for OEMs to create extensive charging infrastructure across India will boost market growth.**

Government initiatives to boost EV manufacturing and sale

By 2030, it is speculated that sales of internal-combustion-engine vehicles will be banned by the government. In this scenario, overall market size of global EVs is projected to augment from US\$ 129,671.56 million in 2018 to US\$ 359,854.56 million by the end of 2025 at a CAGR of 15.69 per cent. Other than technological progress in this sector, additional factors like presence of new mobility modes, increasing shift towards sustainable and energy-efficient modes of transport, increasing government support, and stringent rules and regulations are speculated to strengthen the market of EVs during the projected period.

Investment from foreign firms in the Indian automobile sector has been increased by the government of India, and it permits hundred per cent FDI

under the automatic route. Reports also surfaced that to boost the automobile industry according to global standards, the government is aiming to commence new R&D centres at a total cost of US\$ 388.5 million under National Automotive Testing and R&D Infrastructure Project (NATRIP).

Under FAME-II scheme, around eleven cities have been finalised for the introduction of EVs in their public transport systems by Ministry of Heavy Industries, coupled with incubation centres for startups to research EVs. FAME-II scheme was passed in February 2019 with a fund requirement of US\$ 1.39 billion in next two years.

National Mission for Electric Mobility (NCEM) has launched National Electric Mobility Mission Plan (NEMMP) 2020, intending to invest ₹ 140 billion in the next eight years for the development of electric infrastructure. Government incentives and subsidies and initiatives from OEMs would help reduce the cost of EVs, which, in turn, would drive demand for EVs in the Indian market.

Sale of electric vehicles reached a total of 759,600 units in fiscal 2019. This comprises electric two-wheelers (126,000), electric three-wheelers (630,000) and electric passenger vehicles (3600). The electric two-wheeler market witnessed a year-on-year (YoY) growth of 130 per cent. The industry witnessed a growth due to announcement of FAME-II through which around 100,000 registered electric two-wheelers will be eligible to avail incentive of ₹ 20,000 each, but the catch is the two-wheelers must have ex-factory price within prescribed limits and companies have to register themselves under FAME to avail the scheme. The government will provide an incentive of ₹ 15,000 to 35,000 electric four-wheelers with an ex-factory price of up to ₹ 1,500,000 and incentive of ₹ 13,000 each to 20,000 strong hybrid four-wheelers with an ex-factory price of up to ₹ 1,500,000.

Increasing EV-related training courses in colleges will boost EV R&D in India. Significant advances in financial support from government in the form of tax rebates, subsidies and regulations to promote eco-friendly vehicles will further increase adoption of EVs in the Indian market. **EFY**